DRISHTI AI — Executive Brief

Intelligent Edge Driver Monitoring & Hybrid Satellite Telematics for Heavy Commercial Fleets

Author & Lead Architect:
Rohit Rajesh
Commercial Vehicle Prototype | AIS-140 Ready

1. THE BIG IDEA

DRISHTI AI (*Driver Risk Identification & Safety Telematics Intelligence*) is an automotive-grade, edge-computed Driver Monitoring System (DMS) and telematics safety gateway for heavy commercial transport (multi-axle freight trucks, tractor-trailers, long-haul buses). It transforms standard vehicular cockpits into an active safety environment that continuously monitors driver fatigue and distraction, dynamically modulates risk thresholds using vehicle CAN-bus road speed, and guarantees **zero-loss safety connectivity** in remote cellular dead-zones via automated failover to **JioSpaceFiber (SES MEO satellite backhaul)**.

2. THE PROBLEM STATEMENT (THE FREIGHT CORRIDOR CRISIS)

- **Fatal Highway Micro-Sleeps:** Driver fatigue accounts for over **40% of fatal heavy commercial vehicle crashes** on Indian freight expressways (NH-44, NH-48). At 80 km/h, a 1.5-second micro-sleep leaves a 40-tonne tractor-trailer unguided across 33+ meters.
- **The 'Blackout Corridor' Blindspot:** Critical freight passages (Western Ghats, mining corridors, mountain bypasses) suffer severe 4G/5G cellular dropouts. Traditional fleet telematics go completely dark precisely when accident risk is highest.
- **Cloud Latency & Satcom Cost Traps:** Conventional camera telematics attempt to stream raw video to the cloud for inference, incurring prohibitive satellite bandwidth costs and multi-second latencies incompatible with real-time collision prevention.

3. THE PROPOSED SOLUTION: EDGE INTELLIGENCE & RESILIENT SATCOM

Dual-Model Edge AI Vision

Combines real-time **MediaPipe 468-point FaceMesh** (computing EAR, MAR yawning, and 3D head pose at 30 FPS locally) cross-validated by a **Fine-Tuned Hugging Face Transformer** to eliminate false alarms from glasses or shadows.

Speed-Scaled Dynamic Risk Governor

CAN-bus telematics dynamically scale risk: at 0 km/h (toll gate), looking away triggers no alarm; at 80+ km/h (highway cruise), a 1.5s eye closure instantly triggers high-decibel acoustic in-cab buzzers and brake escalation.

Hybrid JioSpaceFiber Satellite Uplink

When cellular drops, DRISHTI AI bypasses video streaming and transmits an **ultra-compact 48-byte telemetry burst** (DRST|V1|R:HIGH|F:DROWSY|SPD:82) over SES MEO satellite, consuming <1 KB per incident with 100% all-India coverage.

Dual-Screen Operational Command

Laptop/Cab: Full digital truck cluster (analog speedometer, RPM, dual air brake PSI, audio buzzers).
Evaluator Phone: Scans QR code to become live Fleet HQ with haptic vibrations and remote buzzer wake-up buttons.

4. WHAT I AM PLANNING TO DEVELOP (ROADMAP & NEXT STEPS)

- **Phase 1: Automotive Embedded Edge Porting:** Migrate pipeline to automotive-grade edge compute (NVIDIA Jetson Orin Nano / NXP i.MX8) paired with wide-angle Near-Infrared (NIR 940nm) cameras and active IR illumination for robust night driving in pitch-black cabs.
- **Phase 2: Physical SAE J1939 CAN-Bus Interfacing:** Build an isolated physical OBD-II / CAN transceiver interface compliant with AIS-140 and ISO 26262 functional safety, capturing raw commercial vehicle steering angles, pneumatic air-brake reservoirs, and retarder telemetry.
- **Phase 3: JioSpaceFiber Satcom Terminal Integration:** Formalize technical integration testing with Reliance Jio Enterprise / Jio Space Technology to interface the 48-byte telemetry packet with commercial vehicular satellite terminals (flat-panel Phased Array antennas).
- **Phase 4: Commercial Fleet Pilot Trials:** Deploy evaluation units across long-haul commercial tractor-trailers (BharatBenz / Tata Prima / Ashok Leyland) on high-fatigue freight corridors (Mumbai–Delhi NH-48) to empirically validate accident-reduction rates.